CS5487 Handout

Differences between KDE and GMM

Antoni Chan
Department of Computer Science
City University of Hong Kong

Here is a quick comparison between KDE and GMM.

	KDE	GMM
number of components	N (number of data points)	$K \ll N$
training	none.	iterative (EM), takes some time.
data usage	uses all data (degrees-of-freedom) to get the center points, bandwidth is hand-selected.	learns a small set of centers from data.
bias/variance	controlled by bandwidth h: unbiased only if $h = 0$, but has large variance.	asymptotically unbiased (MLE).
storage	needs all the training points.	summarized by a few parameters.
likelihood calculations	kernel values for N components (slower).	likelihoods for $K \ll N$ components (faster).
interpretability	difficult to interpret.	some physical interpretation (clusters, subgroups).
key parameter	picking h is crucial.	picking K is crucial.